AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

Fin340 and 340-F

Course Title: ASSET PRICING AND CORPORATE FINANCE
Version A

Final Examination
2024-2025 / Spring

16-June-2025

Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examinatio
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

Solve All questions

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- **Question 1: (25 marks)**

A public company has 700,000 in preferred stocks with a dividend of 7.2 percent and a current price of $90 a share, and 3 million shares of common stock outstanding selling for $70 a share. The common stock has a beta of 1.40. The company has 65,000 bonds outstanding that are selling for $1,000, bonds with similar characteristics are yielding 7.7 percent. The U.S. Treasury bill is yielding 4.75 percent and the return on the market is 18 percent. The corporate tax rate is 35 percent. What is the firm's weighted average cost of capital?

- **Question 2: (20 marks)**

If a company's EBIT is to be $500,000 every year forever. It can borrow at 22.5 percent. The company cost of capital is 27 percent and has no debt. The tax rate is 35 percent. Bruce will borrow $100,000 and use the proceeds to repurchase shares. What will the WACC be after recapitalization? What do you conclude from the WACC you calculated?

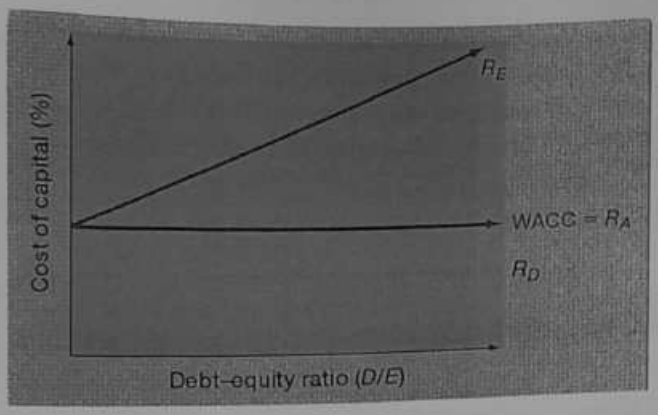
- **Question 3: (25 marks)**

A client plans to send a child to college for four years starting 18 years from now. Having set aside money for tuition, she decides to plan for room and board also. She estimates these costs at $20,000 per year, payable at the beginning of each year, by the time her child goes to college. If she starts next year and makes 17 payments into a savings account paying 22 percent annually, what annual payments must she make?

- **Question 4: (15 marks)**

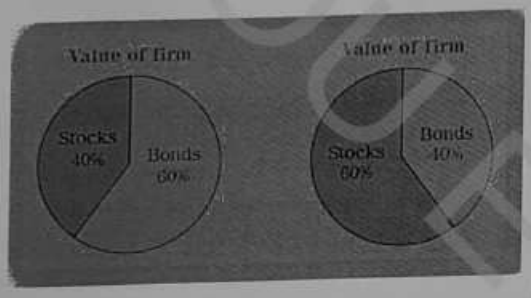
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Explain the figure below:



- **Question 5: (15 marks)**

Explain the below graph:



End of Questions

Formulas

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Present value: $PV = FV / (1 + r)^t$

Future value: $FV = PV \times (1 + r)^t$

$$\text{Annuity present value} = C \times \left(\frac{1 - \text{Present value factor}}{r}\right)$$

$$= C \times \left(\frac{1 - [1/(1+r)^n]}{r}\right)$$

$$\text{Annuity FV factor} = (\text{Future value factor} - 1)/r$$

$$= [(1+r)^n - 1]/r$$

Value of the unlevered firm= $V_U = \frac{EBIT - Taxes}{R_U} = \frac{EBIT \times (1 - T_C)}{R_U}$

Value of the levered firm= $V_L = V_U + T_C \times D$

$V_E = V_L - D$

The cost of equity= $R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$

Cost of equity: $R_E = R_A + (R_A - R_D) \times (D/E)$

The cost of equity using the dividend growth model approach: $R_e = D_1 / P_0 + g$

Cost of Preferred Stock= D/Po

Weighted average cost of capital= $WACC = (E/V) \times R_e + (D/V) \times R_D$

Weighted Average Cost of Capital (WACC): $(E/V) \times R_E + (P/V) \times R_P + (D/V) \times R_D \times (1 - T_C)$, where R_P is the cost of preferred stock.

Weighted Average cost of capital= $WACC = (E/V) \times R_E + (D/V) \times R_D \times (1 - T_C)$

Debt-equity ratio= total debt/total equity

Security Market Line (SML) Approach: $R_E = R_f + \beta_E(E(R_M) - R_f)$

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